Tomato Leaf Disease Detection Using Landing Lens

Smarth Kaushal || 3rd Year CSE

102497023 @ TIET

skaushal1_be23@thapar.edu

62395-12412

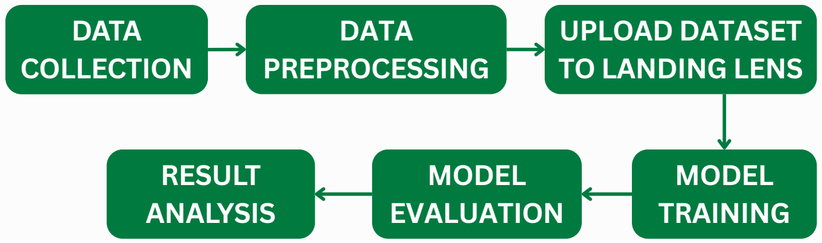
1. Abstract

This project leverages LandingLens, an AI-powered computer vision platform, to detect and classify tomato leaves into 10 categories — **9 distinct diseases and 1 healthy class**. By automating the diagnosis process, it enables early intervention, reduces crop losses, and supports precision agriculture, providing farmers with a powerful tool for real-time, on-field disease detection.

2. Introduction

Tomato cultivation is highly susceptible to multiple leaf diseases, making early detection crucial for yield protection. This project employs AI-based image classification via LandingLens, integrated into a mobile-accessible platform, to deliver quick and accurate disease identification directly in the field.

3. Methodology



4. Input / Output

Test Image	Actual	Predicted	Result
	Bacterial Spot	Bacterial Spot	✓
	Leaf Mold	Leaf Mold	✓
	Healthy	Healthy	✓
	Early Blight	Early Blight	✓

5. Result

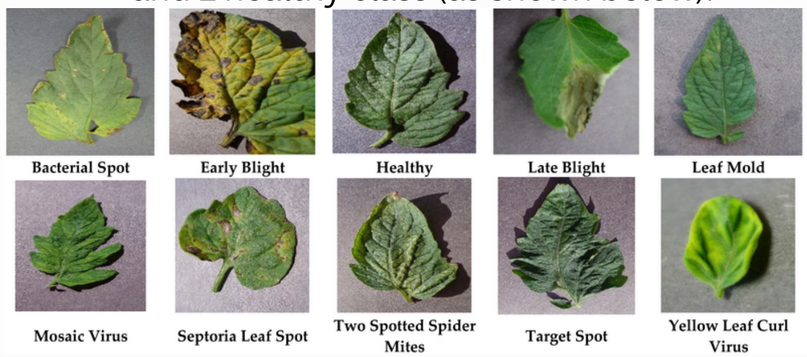
The model attained an accuracy of 100% on the training set, 97% on the validation set, and 88% on the test set, indicating excellent learning performance with strong generalization to unseen data.

6. Conclusion

The system accurately detects and classifies tomato leaf diseases, enabling timely intervention to reduce crop loss and improve yield. Its real-time deployment via mobile or drone platforms supports precision farming and sustainable agriculture.

7. Steps to use the Application

Step-1. Collect Sample Photos – Save some images from 10 categories: 9 tomato leaf diseases and 1 healthy class (as shown below).



Step-2. Scan the QR code.

Step-3. Choose Any Photo – Select a sample image from the gallery.

Step-4. Click "Use Photo" and Get instant disease classification results.

8. References

1. Mohanty, S.P., Hughes, D.P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. DOI Link
2. Ferentinos, K.P. (2018). Deep learning models for plant disease detection and diagnosis. View Paper
3. Landing AI – LandingLens Official Website
4. Kaggle - PlantDisease Dataset. Kaggle Dataset